

UNITEDHEALTH GROUP

Predictive Analytics

Lunch & Learn

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August 2026



The SOA Predictive Analytics Program

Where to build these skills formally

Two SOA e-learning programs cover the full modelling workflow, from framing a problem to validating a model.

[LEARN MORE](#)

[Program overview](#)

[soa.org/programs/
predictive-analytics-certificate](https://soa.org/programs/predictive-analytics-certificate)

1

Predictive Analytics Foundations

[Eight modules](#) · [the core workflow](#)

Problem definition and project management • Data sources, structures, and manipulation • Data exploration and visualization • Feature generation • Feature selection • Model development and validation • Further topics in model selection and assessment • Further topics in decision trees and generalized linear models

2

Advanced Predictive Analytics

[Four modules](#) · [beyond the basics](#)

Additional model families, more techniques in model evaluation and explanation, and an ethical framework for using data and models.

3

What changed for 2026

[New format](#) · [assessment retired](#)

Since January 2026 all Predictive Analytics e-learning modules sit inside [PD Edge+](#), the SOA's subscription platform, available as an individual or corporate subscription that also includes live webcasts, courses, and 1,000+ on-demand items. Module content is unchanged. The final assessment is no longer offered — the November 20–24, 2025 window was the last — and the certificates themselves ended after November 8, 2025.

Session objectives

What you'll leave with

- 1 Know when predictive analytics is worth pursuing
- 2 Ask the questions that reveal a weak project or model
- 3 Understand what leaders still own after deployment

Where we are going

- 1 **Frame** What predictive analytics is, and where it fits in actuarial work
- 2 **Decide** Whether a given project is worth doing
- 3 **Build** Data, features, and model choice
- 4 **Prove** Validation, metrics, and baselines
- 5 **Govern** Monitoring, ethics, and accountability

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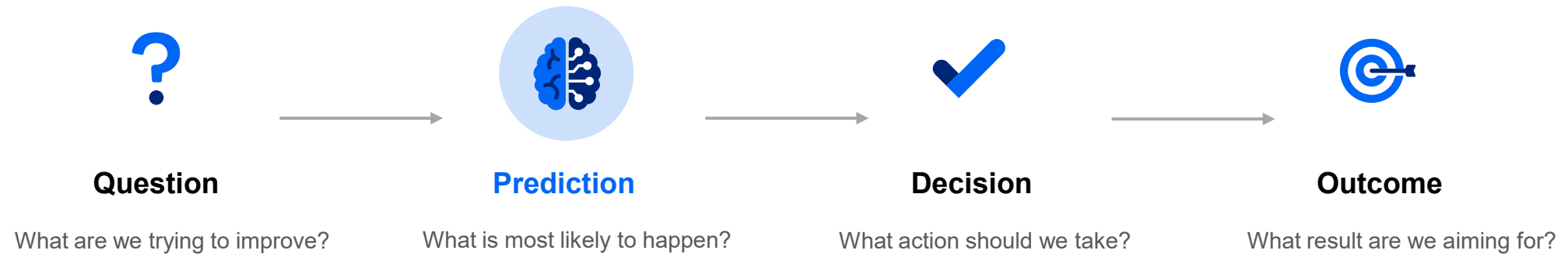
**A prediction is only worth what someone does
differently because of it.**

Frame

1 · What predictive analytics is, and where it fits in actuarial work

Every great decision starts with a **prediction**.

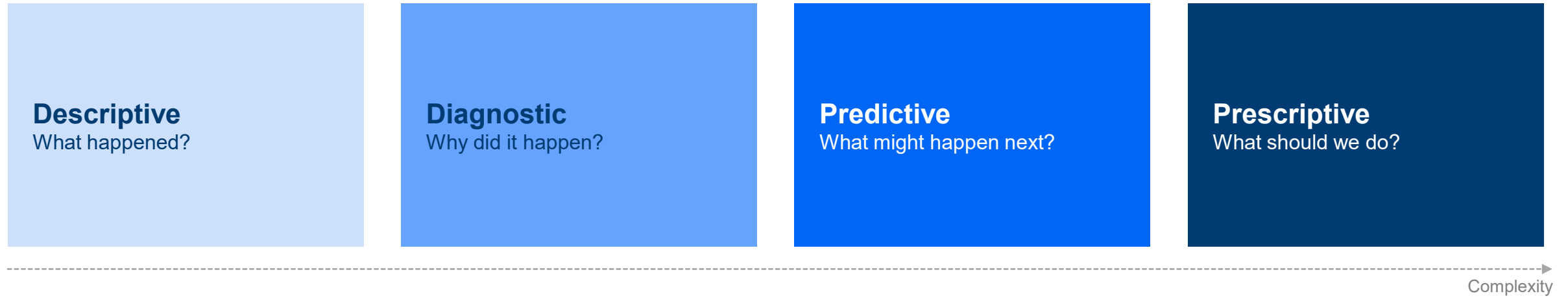
Every day, we make decisions about the future. Before we act, we form a view of what is most likely to happen. Decisions follow a similar thought process:



Predictive analytics helps make the prediction step more structured, so decisions can be made with greater confidence.

What can a prediction tell us?

Four types of analyses; we'll focus on predictive analytics.



We do this intuitively every day. Machine learning does it with data, spotting patterns we cannot see, so when is that worth doing?

Prediction is not the product, better decisions are.

So what does predictive analytics actually do?

It uses what already happened to estimate what happens next.

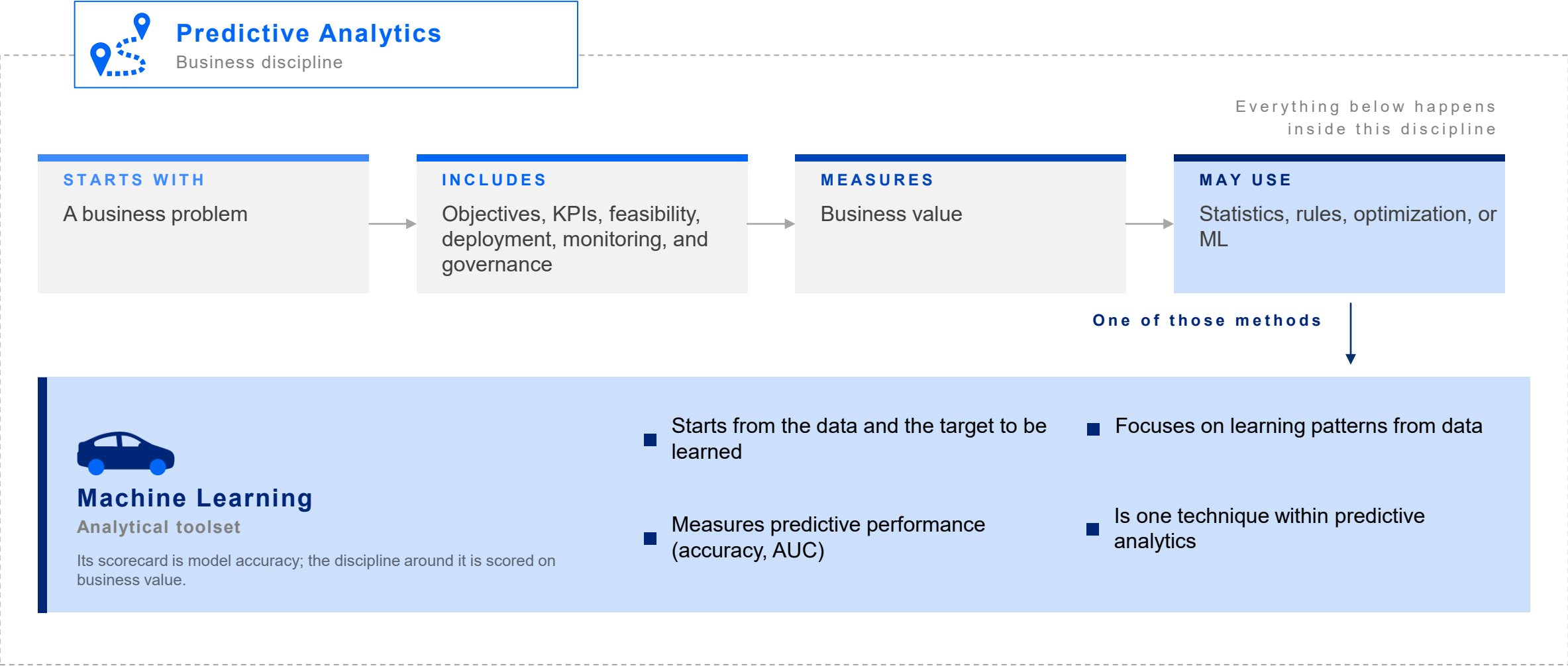
It helps answer

- 1 What might happen next?**
The core question behind every model.
- 2 Who should we pay attention to first?**
Ranking, not certainty.
- 3 How likely is an event?**
A probability, with a time window.
- 4 Where should limited resources go?**
The reason the ranking matters.

It does not

- ✗ Eliminate uncertainty**
It sizes the uncertainty; it does not remove it.
- ✗ Prove cause and effect**
A pattern is not a reason.
- ✗ Make decisions on its own**
People still choose, and still own the choice.

Predictive Analytics Is the Journey. ML Is One Vehicle.



When is ML the right tool?

ML creates the most value when predictions can drive meaningful action.

Works well when...

1

Clear Decision to support

The prediction should influence a specific action, such as prioritizing outreach, allocating resources, or identifying risk.

2

Historical data is available

ML learns from past examples, so reliable historical data is needed to recognize patterns.

3

Patterns are difficult to identify manually

When relationships involve many variables and interactions, ML can uncover patterns that are hard to detect through rules alone.

4

Better predictions lead to action

The value of a model comes from changing decisions or behaviors, not simply generating more accurate predictions.

May not be needed when...

1

Rules already solve the problem

If a simple business rule achieves the desired outcome, adding machine learning may introduce unnecessary complexity.

2

Little or no reliable data exists

Without sufficient high-quality data, model predictions are unlikely to be reliable or actionable.

3

Predictions won't change decisions

If stakeholders will make the same choice regardless of the prediction, the model provides limited business value.

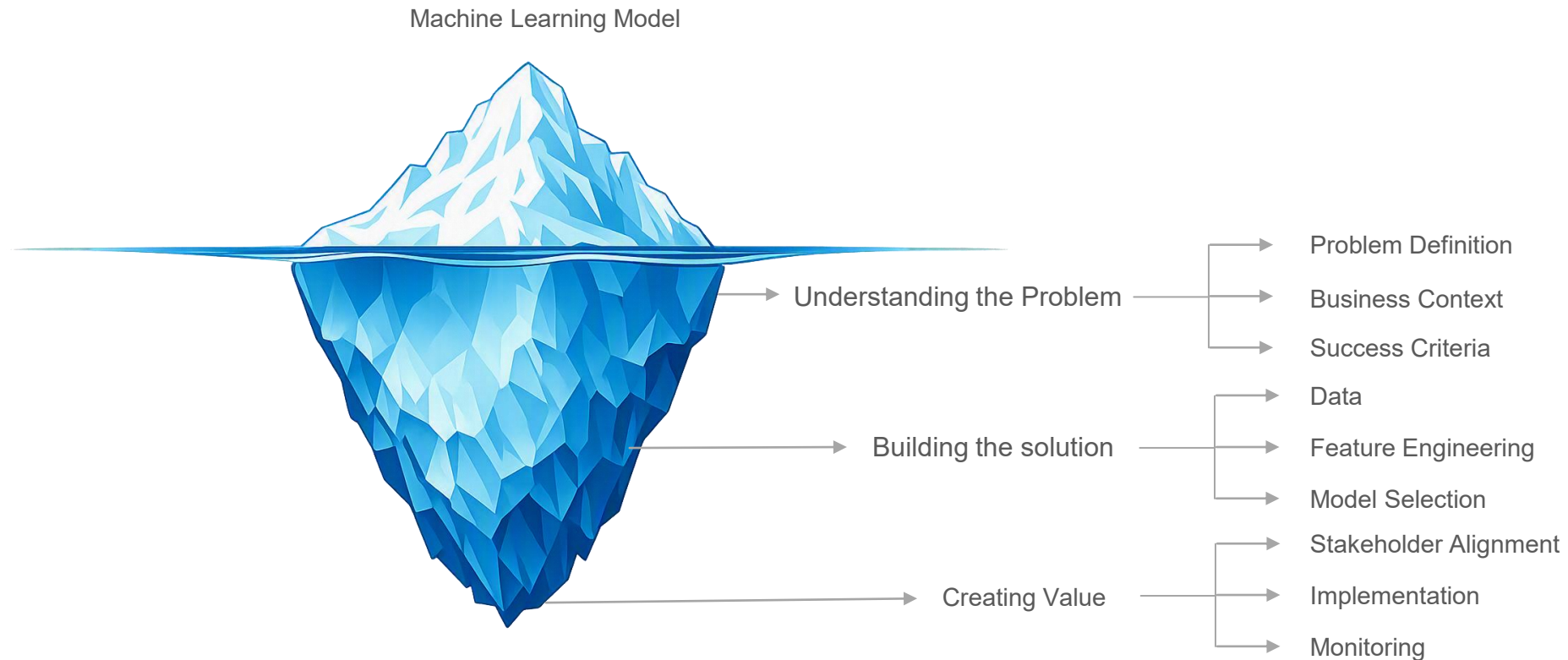
4

Simpler approaches perform just as well

When basic rules or traditional analytics achieve similar results, the simpler solution is often easier to maintain and explain.

An ML Model Is Only Part of the Story

It is a much larger process than just selecting the right model. The success of an ML project depends far more on understanding the problem.



Where does actuarial judgment fit?

Predictive analytics doesn't replace actuarial judgment. It gives that judgment better information.

The workflow is familiar

Define the Problem -> Design a Solution -> Monitor Results
(Actuarial Control Cycle)

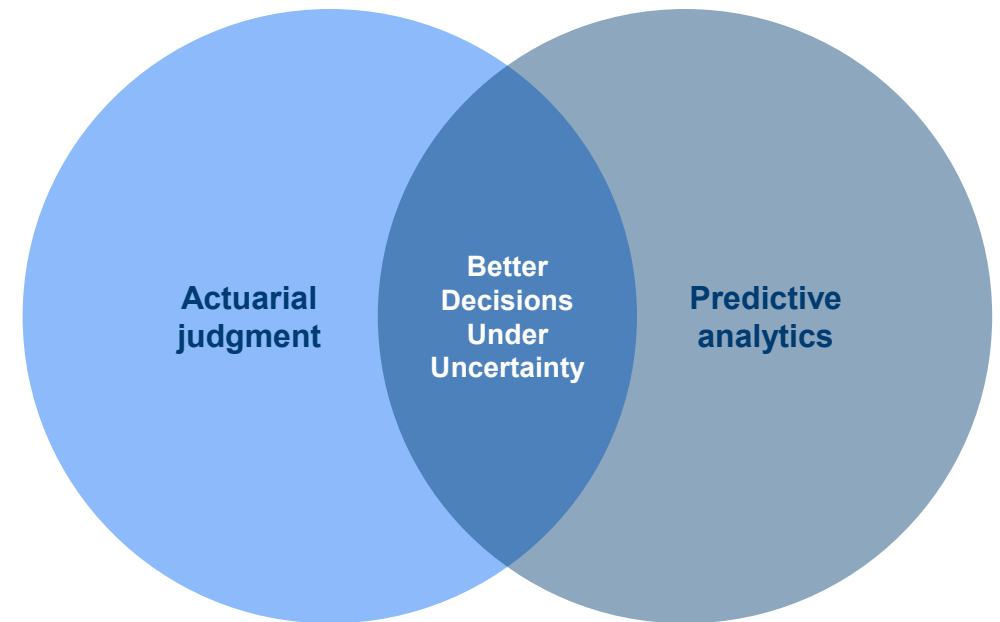


Scale is what changes

More data, and patterns that are hard to see with traditional methods.

Judgment still decides

The model informs the decision; business context and actuarial judgment make it.

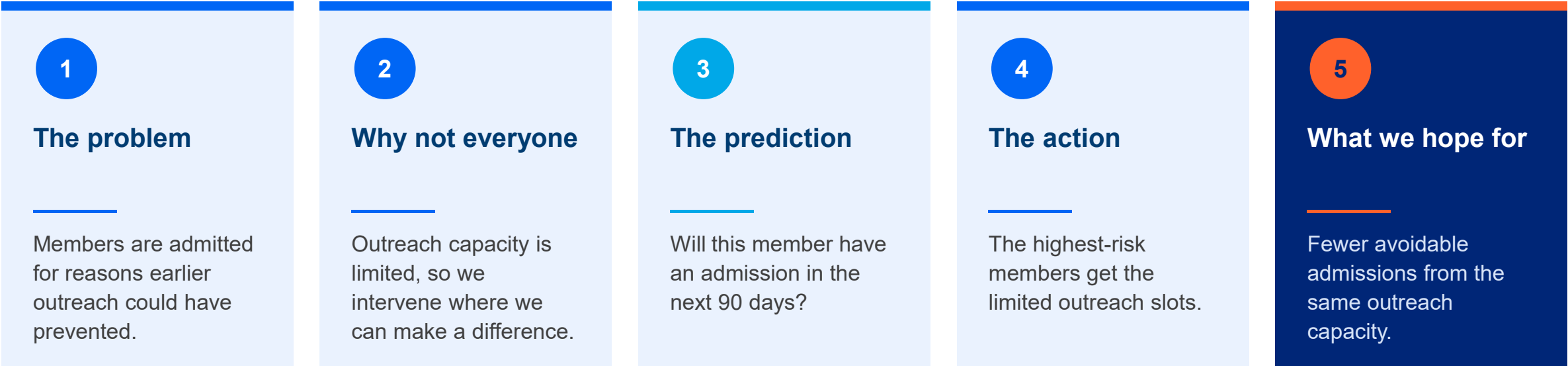


Evidence sharpens judgment; judgment still decides.

Our example case study: Avoidable Admissions

We'll come back to this one case as each idea in the session comes up.

THE CASE



HOW TO FOLLOW IT: an orange line at the bottom of a slide is this case study coming back.

Case study: the decision is who gets care management outreach.

Decide

2 · Whether a given project is worth doing

What should every leader ask?

Six questions decide whether a project is worth doing, and whether its answers can be trusted.

1	What decision changes?	Name the decision it improves, and the value follows.
2	Better than what baseline?	Every model has to beat a named alternative.
3	Do we trust the data, and do we know the business?	Data we can rely on, and the expertise to turn it into a model that makes sense.
4	What happens if we are wrong, or unfair?	Errors cost different amounts, and they do not land evenly on people.
5	How will the prediction be used?	Good models fail when the workflow never changes.
6	Who owns monitoring and change?	It keeps scoring after launch, and the world moves under it.

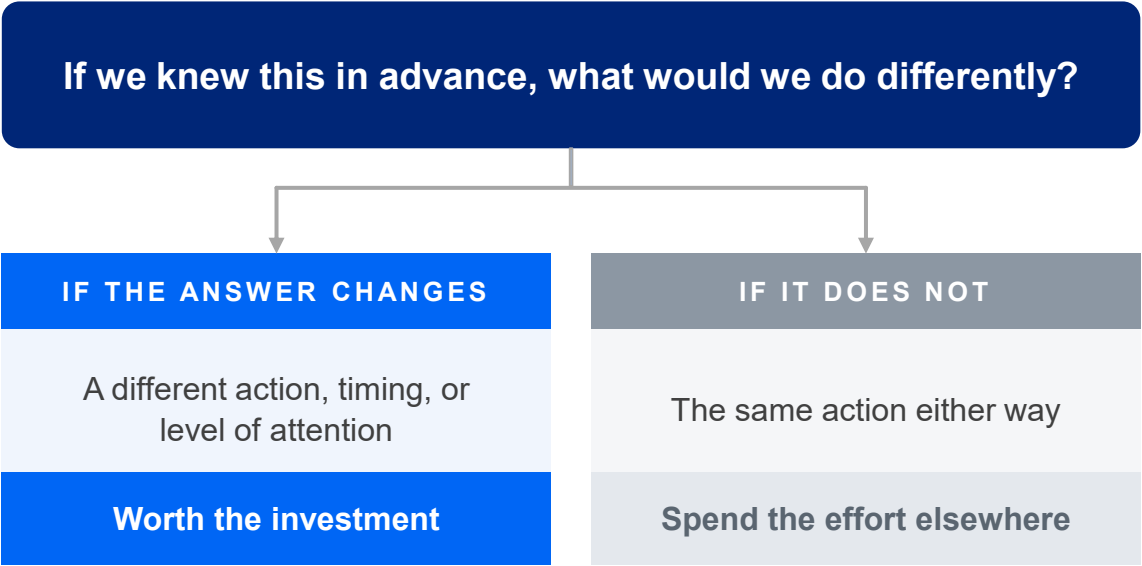
What decision changes?

The first question is not 'Which Model?'
A prediction has no value unless it changes a decision.

What to know

- | | |
|---|--|
| 1 | Name the decision and who makes it. |
| 2 | Describe today's process and where it falls short. |
| 3 | Name the action the prediction enables. |

The test for any prediction



Don't build a model without clarity of purpose.

Case study: the decision is who gets care management outreach.

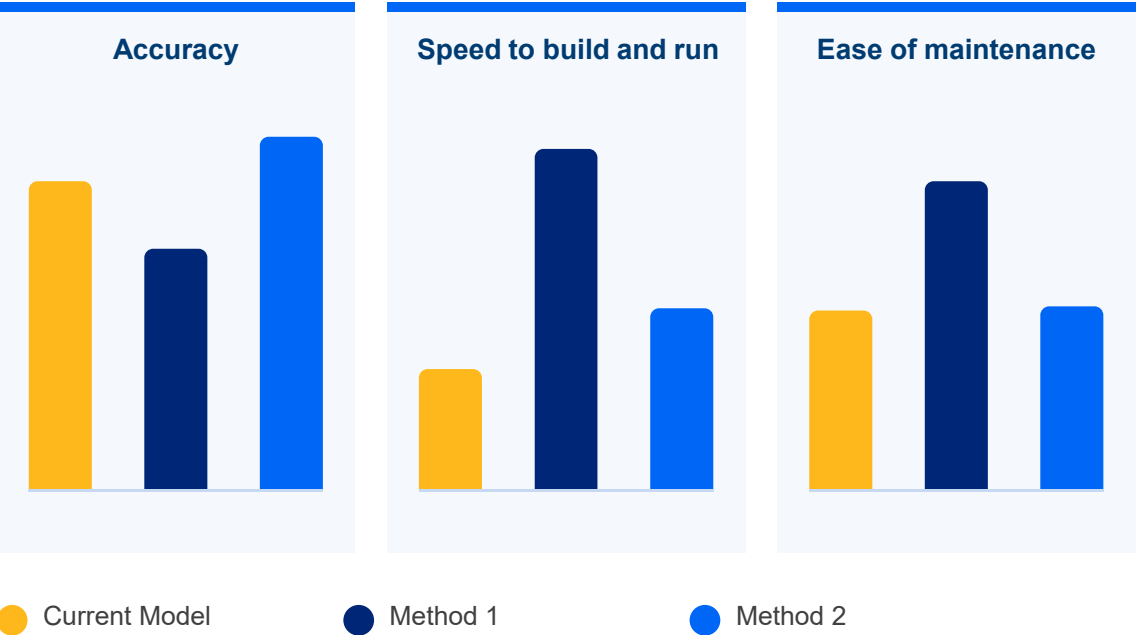
Better than what baseline?

Every model competes against something.

What to know

- 1 Models should beat the current approach.
- 2 Compare on the same population.
- 3 Compare on the same time period.
- 4 Translate improvement into operational value.

Each option trades one strength for another



Never compare a model to another model before comparing it to today's reality.

Do we trust the data, and do we know the business?

A model needs data we can rely on, and people who understand the business that data describes.

What to ask

- 1

Is the data complete, accurate, and consistent across sources?
- 2

Is it representative of the population we will actually score?
- 3

Are the right subject-matter experts in the room while we build it?
- 4

Would an expert recognize these as the drivers that matter?
- 5

Could any feature leak information we would not have at the moment of prediction?

What good looks like



Data quality

Sources understood, gaps known, errors found before launch.



Representativeness

Selection and measurement bias tested, not assumed.



Business knowledge

The people who know the process shape what the model sees.



Features that make sense

Every variable is explainable, and defensible.

Detailed slides on data and features follow in the Build section.

A model cannot make sense without trustworthy data and business knowledge.

Case study: care managers know prior admissions, medication burden, and ER visits are the signals that matter.

What happens if we are wrong, or unfair?

Not all errors are equally expensive, and they do not land evenly on everyone.

What to ask

1	Which mistake hurts most?
2	Missed opportunity, or wasted intervention?
3	Who is harmed, and are some groups harmed more?
4	Fairness, regulatory, and reputational risk?

Which error costs more here?

	Actually happens	Does not happen
We predict it	Correct call	False alarm: we spend effort where it was not needed
We do not	Missed case: someone who needed help gets none	Correct pass

The two highlighted cells carry the real cost. Which one is worse here?

Answer both: what the mistakes cost, and who bears them.

Because the event is rare, the model must be built and measured for imbalance, or it will quietly optimize for the majority “no” case.

Case study: a false alarm wastes a call; a miss leaves a high-risk member without help — so we check whether misses cluster in one group.

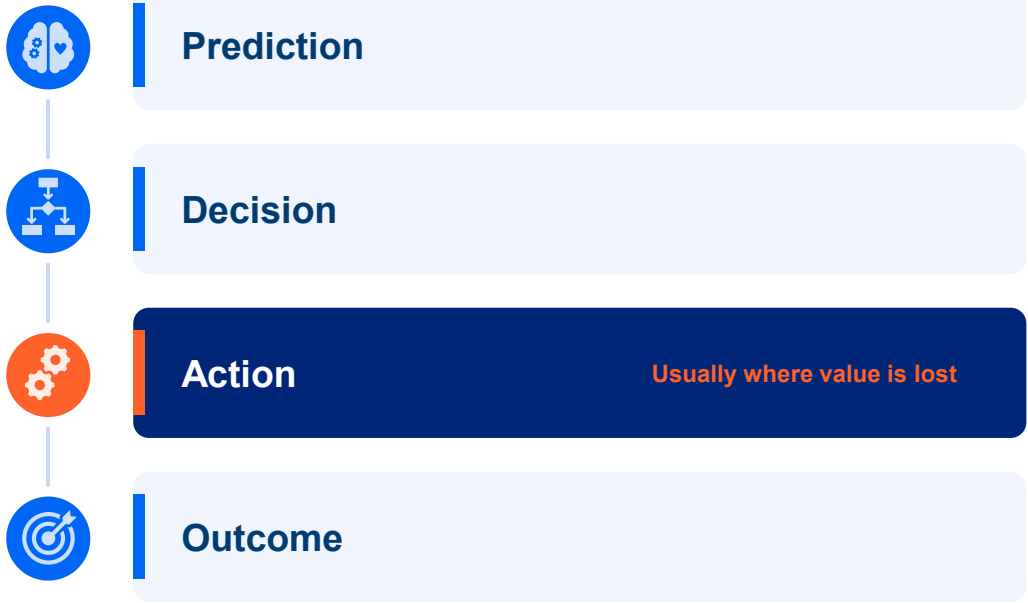
How will the prediction be used?

Many technically successful models create no value because nobody changes the workflow. Predictions create value only when workflows change.

What to ask

- | | |
|---|-------------------------------|
| 1 | Where does the score appear? |
| 2 | Who sees it? |
| 3 | What do they do differently? |
| 4 | Is action capacity available? |
| 5 | How is value measured? |

The path to value

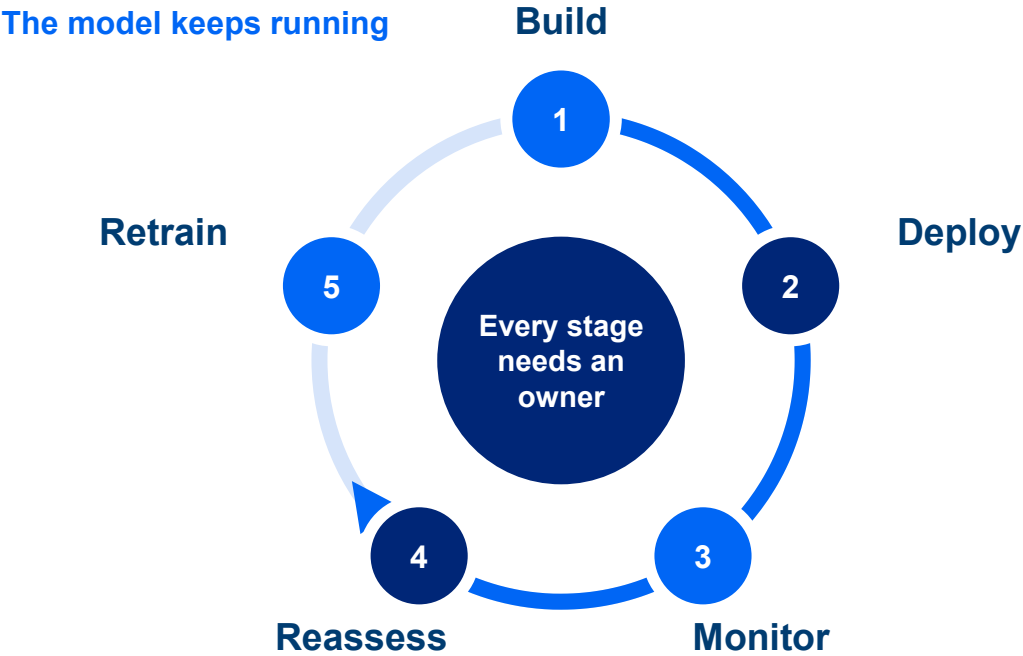


Who owns monitoring and change?

Models continue making decisions after deployment. Models inform decisions. People remain accountable.

What to ask

1	Who owns the model?
2	Who monitors drift?
3	Who receives alerts?
4	Who approves changes?
5	When is retirement required?

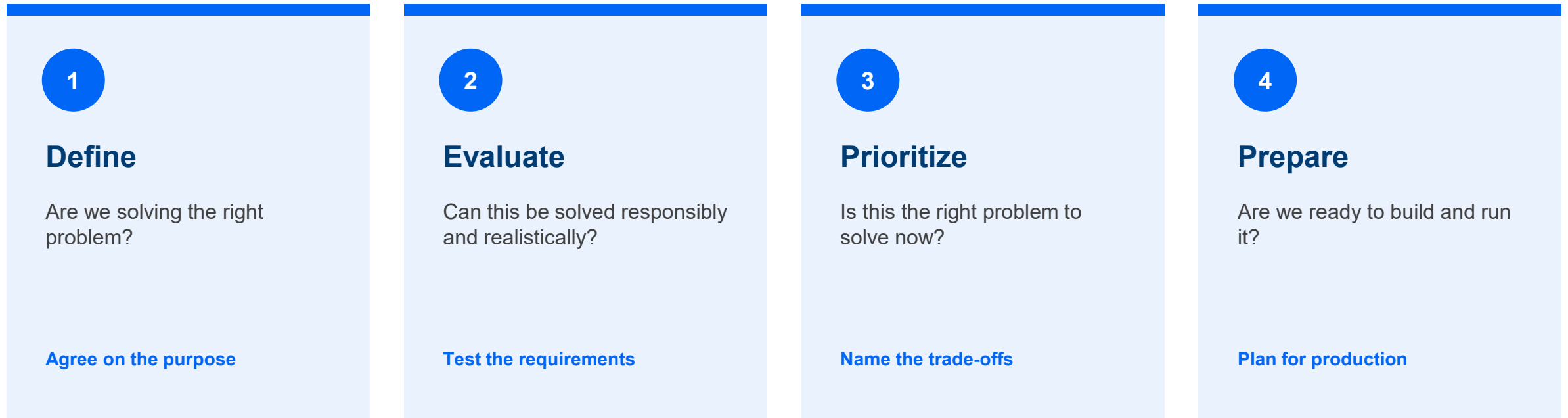


Build

3 · The project lifecycle, data, and features

How leaders keep ML projects on track

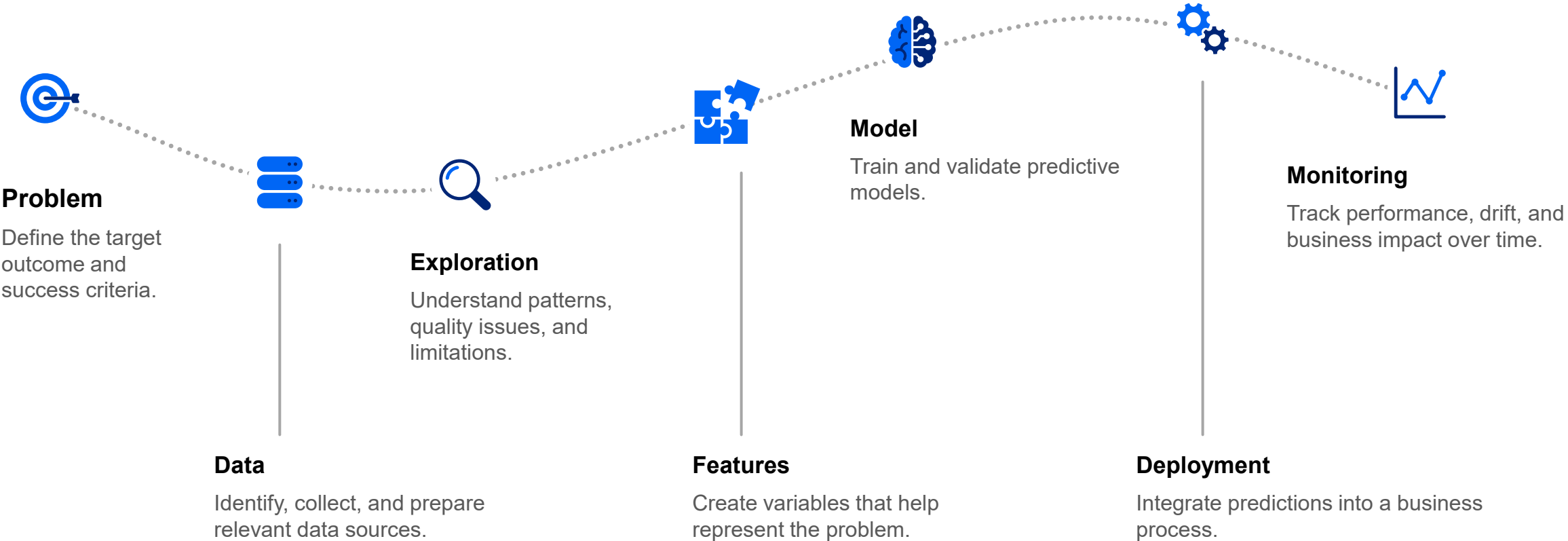
The SOA framework for ML project management



Project outcomes are often influenced more by early decisions than by model selection.

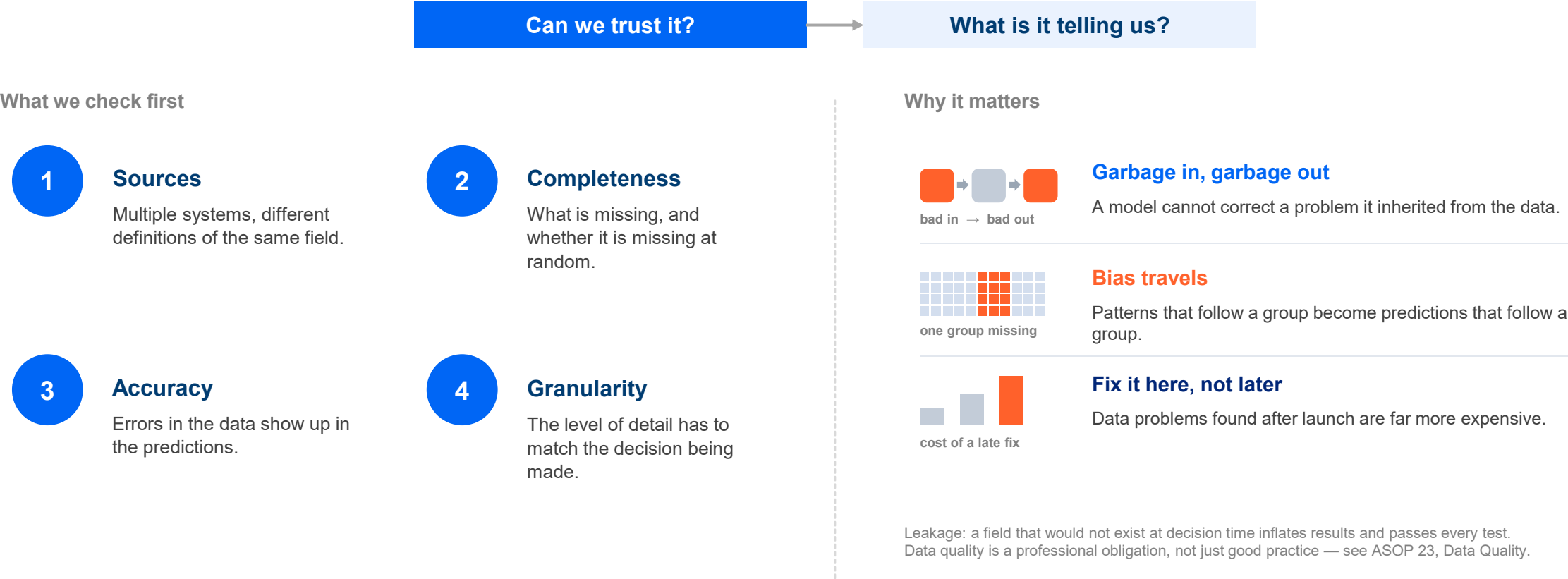
More detailed slides on these concepts are in Appendix A

The lifecycle of an ML project



Can we trust this data?

Before a model can learn anything, the data has to be worth learning from.



What is this data telling us?

Once the data is trustworthy, we look for what it actually says.

Can we trust it?

What is it telling us?

What we look for



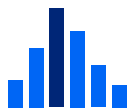
Patterns

Trends and relationships that may carry predictive signal.



Outliers

A data error, or a real event that deserves attention.



Distributions

Skew and imbalance change what a model can learn.



Relationships

Variables rarely act alone. Combinations matter.

Why it matters

Explore before you model

Most of the insight is learned before any algorithm runs.

Judgment is required

Statistics flag the oddity; people decide what it means.

This shapes the features

What we find here becomes what the model can see.

Exploration is where the business question meets the evidence.

Case study: a spike in ER visits is the strongest early signal in the data.

Feature engineering: where judgment enters the model

Features are how actuarial knowledge is written into the model. The model learns from what we choose to give it.

1 Features decide

- The model learns from features, not raw data
- Judgment beats brute force here



2 Actuarial building blocks

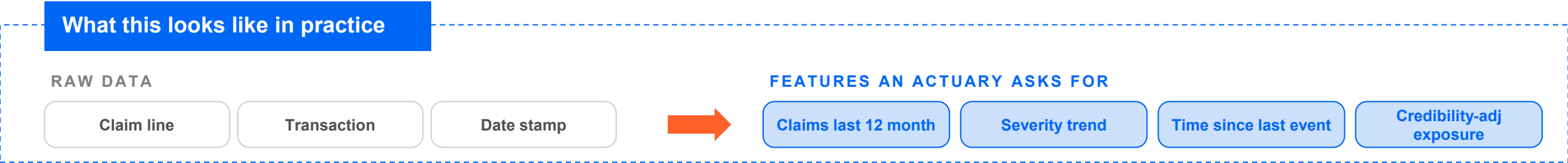
- Frequency and severity
- Trend and duration
- Ratios, lags, and counts
- Exposure and credibility



3 Handle with care

- Group rare categories where signal could be missed
- Cap or transform extreme values
- Use point-in-time values only
- Drop features you cannot explain
- Avoid data leakage — nothing known only after the outcome





Automatic selection methods can drop variables for you, but what they keep is tuned to one model and is not the same as a variable an expert can defend.

This is where actuarial judgment shapes the model. What experts know matters becomes what the model can see.

Case study: prior admissions, medication burden, ER visits, utilization trend.

Build

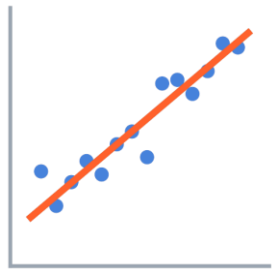
3 · Model selection and trade-offs

The Question Determines The Model

Before anyone compares algorithms, the business question sets the prediction type and the type rules out most of the field.

REGRESSION — PREDICT A NUMBER

How much?



Answer: an amount
Error: distance from the actual value
Value: better estimation
Example: forecast future cost

Cost · Admissions · Duration · Utilization

CLASSIFICATION — PREDICT A CATEGORY

Yes or no, and how likely?



Answer: a probability
Error: false alarm or a miss
Value: better prioritization
Example: who receives outreach first

Admit / not admit · Churn · Fraud · Readmission

The business question determines the prediction type. The prediction type constrains which model families make sense.

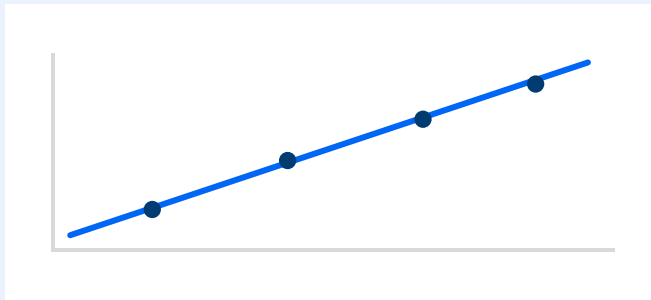
Case study: will this member be admitted in the next 90 days is a classification question.

The Same Problem, Three Different Learning Styles

Each model searches for the signal differently. That is what makes them suited to different problems.

GLM

Looks for consistent relationships. Best when transparency matters.

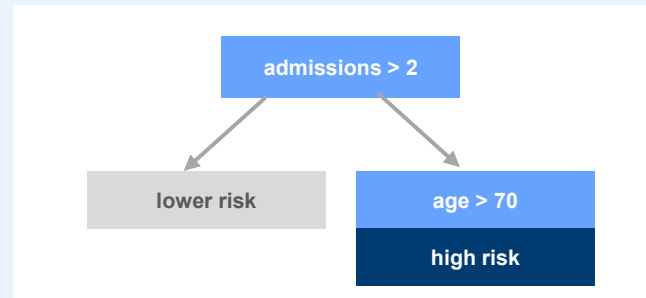


EXAMPLE

When utilization rises, risk rises.

Decision tree

Looks for thresholds and splits. Best when interactions matter.

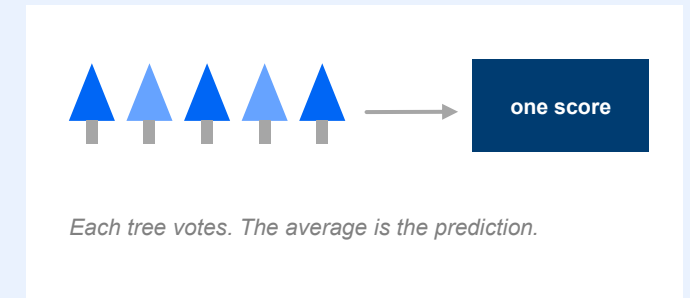


EXAMPLE

If admissions > 2 and age > 70, high risk.

Ensemble

Looks for patterns spread across many relationships. Best when accuracy matters most.



EXAMPLE

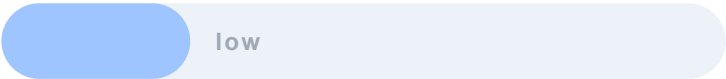
No single factor predicts risk, but many together do.

Complex models see patterns simple ones miss

More capacity means more of the picture. It also means more to explain and maintain.

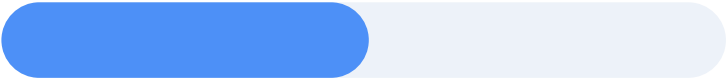
Human judgment

One factor at a time



GLM

Several factors together



Decision tree

Interactions between factors



Ensemble

Many interactions at once



Illustrative. Length shows how much complexity a method can represent, not how accurate it will be.

What the extra capacity costs

Explainability

Harder to justify a single prediction

Governance

More review, documentation, and approval

Monitoring

More to watch after launch

Complexity buys sight. We pay for it in explanation, oversight, and upkeep.

Every Improvement In Performance Comes With An Obligation

Model families are not a horse race. Each one buys something and charges something.

MODEL FAMILY	WHAT IT BUYS	WHAT IT COSTS
GLM COMPLEXITY	Transparency and stability	Less flexibility to capture complex patterns
Decision tree COMPLEXITY	Intuitive segments	Instability; small data changes move the splits
Ensemble COMPLEXITY	Flexibility and predictive power	More explanation, validation, and monitoring

Model selection, use, and reliance are governed by ASOP 56, Modeling. The business need explainability, cadence, regulatory transparency, data maturity, and support, decides which families are eligible.

Complexity has a cost: Would you take 3% more accuracy if it doubled governance effort?

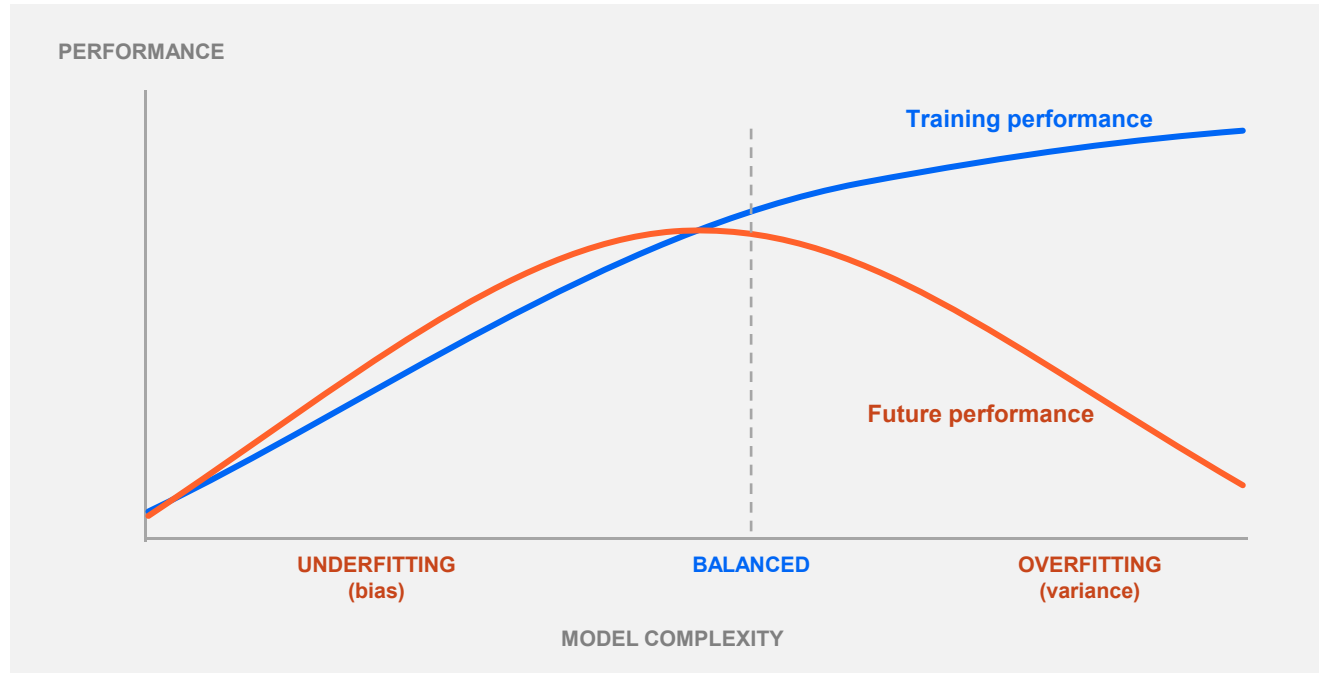
Case study: GLM explains the drivers. Random forest predicts better.

Prove

4 · Validation, metrics, and baselines

A model that explains the past perfectly may fail next year

Fit improves with complexity; performance on data the model has never seen does not.

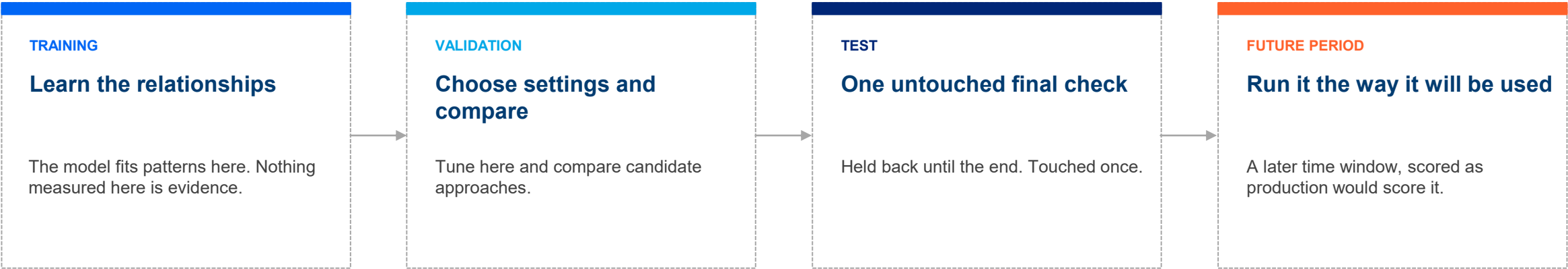


What leaders should ask

- 1 Was this performance measured on data the model had not seen?
- 2 Was the test population comparable to the intended population?
- 3 Does performance hold in a later time period?

Each dataset has one job

Performance is only evidence when the data behind it was kept out of model development.



Cross-validation

Split the data several ways and rerun the test on each split.

Checks whether performance is stable across different samples, not a lucky split.

Cross-validation in practice



Each round holds out a different slice, then rotates.

Where the claim breaks down

Leakage, reusing the holdout, and tuning until the number looks good. Reuse turns a test set into part of development.

Split by group, not by row

With longitudinal claims, keep all of one member's records in a single partition. A member split across train and validation leaks their own history into the score.

There is no single “accuracy” number

Start from the business question. The measure follows from what the decision needs and what mistakes cost.

THE BUSINESS QUESTION	USEFUL MEASURE
How far off are the predicted amounts?	RMSE / MAE · regression
Are the highest-risk members near the top of the list?	Lift / AUC · ranking
Of those we selected, how many truly need action?	Precision · classification
Of those who need action, how many did we find?	Recall · classification
Are the predicted risk levels believable?	Calibration · classification

Why “accuracy” misleads here

Avoidable admissions are rare, so a model that predicts “no admission” for everyone looks about 95% accurate and finds no one. In one course example a model did exactly that across 150,000 records, AUC 0.51, no better than a coin flip and rebalancing the data lifted it to 0.75. That is class imbalance, and it is why we judge on recall and lift.

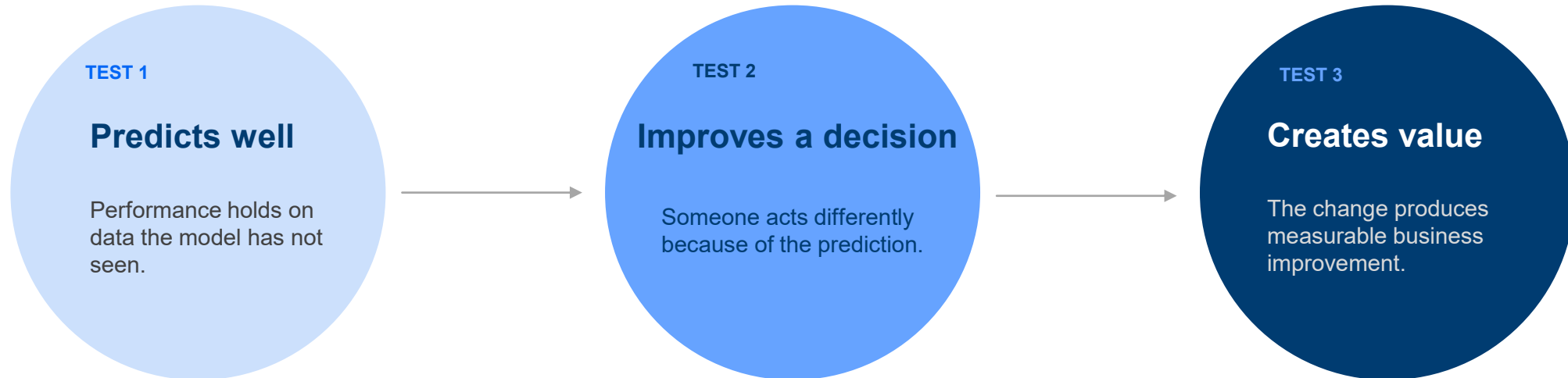
Choose the metric from the decision and the cost of mistakes, not from habit.

Estimation questions call for regression measures; prioritization questions call for ranking and classification measures.

Case study: recall matters more than precision. Missing a member is the expensive error.

A good model passes three tests

Most organizations stop at the first test.



The highest standard is not predictive performance. The highest standard is realized value.

So how do we know the result is good enough?

There is no universal passing score.

WEAK EVIDENCE



What we cannot accept

- Trained and tested on the same data
- No baseline to beat
- A single performance number
- The metric chosen after the fact

STRONG EVIDENCE



What we should expect

- Held-out validation, touched once
- Compared to the current process
- Metrics tied to the decision
- Improvement stated in operational terms

BEST EVIDENCE

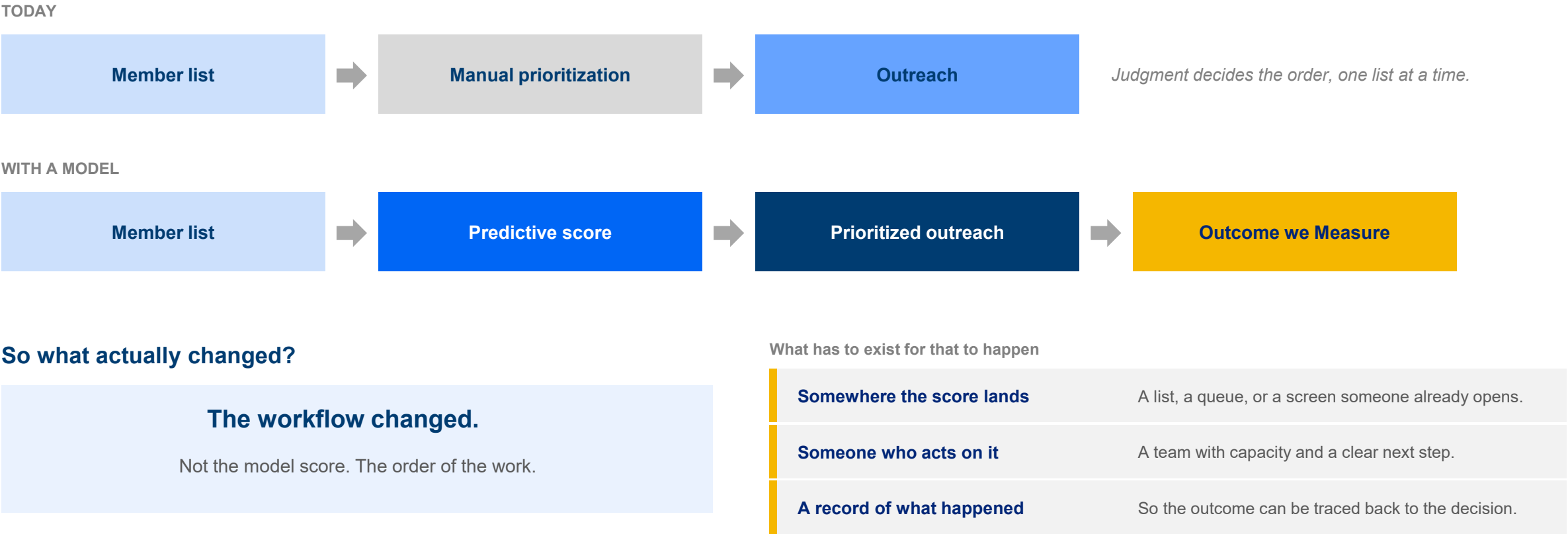


What we should aim for

- Performance holds on a later time period
- The workflow actually changed
- Outcomes moved, not just scores
- Still monitored after launch

A prediction creates no value by itself

The model isn't the deliverable. The changed workflow is.



Case study: the score reorders the outreach list. That is the whole deliverable.

Govern

5 · Monitoring, ethics, and accountability

A model is a live decision system

Models decay and workflows drift. Three things need watching after launch.

LANE 1



Data drift

What the model is being fed changes.

- The population changes
- Coding or source systems change
- Missingness and feature distributions shift

LANE 2



Performance drift

The model still runs, but predictions worsen.

- Ranking weakens
- Calibration changes
- Results vary by population segment

LANE 3



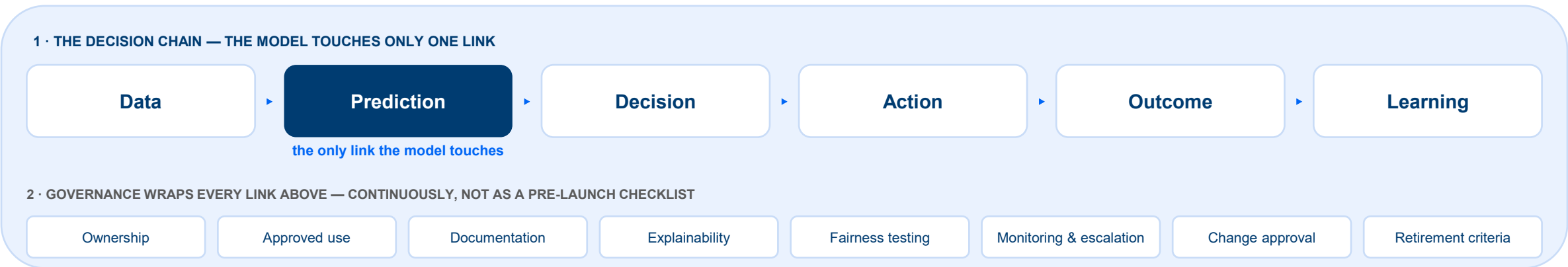
Business drift

The decision around the model changes.

- The workflow stops using the score
- Intervention capacity changes
- Outcomes no longer improve

Models inform decisions. People remain accountable.

The model touches one link. Governance wraps all of them.



3 · WHERE IT BREAKS — EACH FAILURE IS ONE LINK GIVING WAY

BREAKS AT: DATA → PREDICTION

The wrong problem

A model that answers a question no decision depended on.

Communication of results and limitations is governed by ASOP 41.

BREAKS AT: THE WRAP ITSELF

No owner, no explanation

A model no one can explain, and no named person accountable for it.

BREAKS AT: ACTION → OUTCOME

A metric with no value

Impressive performance that never moves a business outcome.

Case study: monitor performance to see whether outreach actually helped.

Governance wraps every link. At each one, a named person — not the model — is accountable.

Accurate is not the same as fair

A model can perform well and still be unsafe, unexplainable, or non-compliant. Leaders own that risk.

Three things every model must be



Fair

No reliance on protected classes or their proxies, such as a ZIP code standing in for race. Test performance across groups, not just overall.



Safe

Reliable, robust, and privacy-protecting. It should fail gracefully and guard member data.



Transparent and accountable

Every prediction can be explained, and a named person owns the outcome.

What leaders must ensure

■ Regulatory footing

Meets applicable ASOPs and privacy law (GDPR, CCPA, HIPAA).

■ No disparate impact

Fairness is tested at the data, model, and implementation stages, not assumed.

■ Documented judgment

Design choices, excluded variables, and known tensions are written down.

■ A tension budget

Fairness definitions are mathematically incompatible: equal selection rates across groups and equal accuracy within them cannot both hold.

Fairness and compliance are design decisions, not a final review step.

Case study: monitor admission-risk scores by age, race, and disability to confirm outreach helps every group, not just the average.

The rules behind the fairness slide

Standards and regulations that govern how these models may be built and used.

Fairness is not one rule. It is a set of standards, statutes, and state-by-state variation that leaders are accountable for knowing.

Actuarial standards of practice

ASOP 12 — Risk Classification

Characteristics used must relate to expected outcomes and be objectively determinable.

ASOP 23 — Data Quality

Disclose sources, reliance, limitations, and any results that are uncertain or potentially biased.

ASOP 41 — Actuarial Communications

Form, clarity, timing, and a named responsible actuary.

ASOP 56 — Modeling

Intended purpose, consistent assumptions, known weaknesses, and reliance on others' models.

Regulation and protected classes

Privacy

GDPR (EU), CCPA, HIPAA, GLBA, FCRA; PIPEDA in Canada.

Protected classes

Race, religion, national origin, age 40+, sex, sexual orientation and gender identity, pregnancy, disability, genetic information.

Proxy risk

ZIP code as a stand-in for race — the reason credit decisions use only the first three digits.

Insurance-specific

Risk classification is lawful discrimination when it reflects actual loss potential; use of sex, age, credit score, and genetics varies by state and by line.

If you remember these five things

Everything in this session reduces to these five ideas — one of which is a set of six questions.

1

Predictive analytics helps estimate what happens next.

It uses what already happened to put a number on what probably will.

2

Prediction is not the product. Better decisions are.

A model that changes nothing is worth nothing.

3

Six questions tell you whether a project is worth it.

Does it change a decision, beat a baseline, rest on data we trust, cost us the right errors, get used, and have an owner?

4

Models change workflows, not outcomes directly.

The value shows up only when the work is done differently.

5

Models decay, and people remain accountable.

Drift is normal. Ownership is not optional.

Appendix A — Keeping an ML Project on Track

Define: What does success look like before we start?

Agree on these before modeling begins, and re-check them whenever the model is reassessed.

		THE QUESTION	ILLUSTRATIVE EXAMPLE
	Decision	What decision changes, and what action becomes possible?	<i>Which members get outreach next month, instead of a broad mailing.</i>
	Target	What exactly are we predicting, for whom, and over what window?	<i>Likelihood of an avoidable admission in the next 90 days.</i>
	KPI	How will success be measured, and what result would make us stop?	<i>Admissions avoided per 100 outreaches, against the cost of the outreach.</i>
	Ownership	Who owns the parts of the process and answers for the result?	<i>The business owner who funds the outreach and acts on the output.</i>

How leaders keep ML projects on track

Evaluate: Can we actually deliver this?

Six feasibility checks decide whether a good idea becomes a working project. Any one of them can stop it.

The feasibility check: settle this before the six leadership questions are worth asking.



Data

Do we have the history, and permission to use it for this purpose?



Authority

Are we allowed to act on the prediction, contractually and legally?



Budget

Is there funding for the build and for running it afterward?



Workflow

Does a path to action exist, or would someone have to invent one?



Technology

Can we score and deliver this at the cadence the decision needs?



Sponsorship

Who has committed to act on the output, and who funds it?

If any answer is no, do not build yet. Fix the issue first.

Prioritize: Is this the right problem to solve now?

Score every candidate the same way, side by side, before any of them is funded.

	Value at stake	How much is the decision worth if we get it right?
	Cost and effort	Rough build cost, plus what it takes to run it every year.
	Project risk	Could the model fail to work at all? Data, signal, timeline.
	Adoption risk	Could it work and still change nothing? Usually the bigger risk.
	Time to first result	Months or quarters? Early wins buy patience for slower work.
	Business priority	Does this move something leadership is already accountable for?

Make it a standing question

Why this project instead of the next-best opportunity?

The score is not the point, the trade-off is.

A weak project that gets funded is usually one that was never compared to anything.

What are we not building this year because we chose this instead?

Prepare: Are we ready to build it and to run it?

Evaluate asked whether it can be done. Prepare names who does what, and what day one in production looks like.



Data pipeline

Not just access. A refresh schedule, quality checks, and a named owner.



Run and support

Who is on call when scoring breaks, and how fast must it be fixed?



Workflow and training

The step changes, and the people doing it are shown how.



Production path

Where the score lands in the system people already work in.



Roles and approvals

Who builds, who reviews, who signs off, and by when?



Decision rights

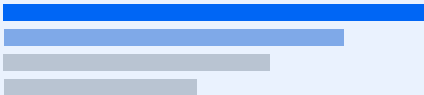
Who acts on the output, and what are they allowed to change?

It is ready when the pipeline, the workflow, and the on-call owner are ready.

Appendix B — ML Models

Four kinds of question, four kinds of problem

The first three predict an outcome. The fourth describes the population instead.

PREDICTS AN OUTCOME Regression <i>How much?</i> Predicts an amount. EXAMPLE: Future cost JUDGED BY: RMSE, MAE 	PREDICTS AN OUTCOME Classification <i>Yes or no?</i> Predicts a category or a probability. EXAMPLE: Admission in 90 days JUDGED BY: AUC, precision, recall 	PREDICTS AN OUTCOME Ranking <i>Who should go first?</i> Orders the population by risk. EXAMPLE: Outreach priority JUDGED BY: Lift, top-decile capture 	DESCRIBES THE POPULATION Clustering <i>Who looks similar?</i> Finds groups; nothing is predicted. EXAMPLE: Member segments JUDGED BY: Stability and usefulness 
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Name the problem type first. It decides the model family, the metric, and what counts as being right.

Predicting a number

Continuous outcomes, where the answer is an amount rather than a yes or no.

EXAMPLES	Future cost	Length of stay	Claim severity	Utilisation next year		
MODEL	WHAT IT DOES	STRENGTHS	WATCH OUT	USE WHEN	EVALUATE WITH	
Linear regression 	Fits a straight-line relationship between drivers and the amount.	Simple, fast, fully transparent.	Assumes effects are straight lines and additive.	Explanation matters more than accuracy.	RMSE, MAE	
Generalised linear model 	Same idea, but fitted to the shape of the data: counts, costs, or rates.	Explainable, and standard in actuarial work.	Requires choosing the right family and link.	Skewed or non-normal outcomes.	RMSE, deviance	
Regression tree 	Splits the population into groups and predicts an amount for each.	Captures interactions; easy to read as rules.	Unstable; small data changes shift the splits.	Segments behave very differently.	RMSE, MAE	
Random forest / boosting 	Combines many trees into one number.	Highest predictive performance.	Hardest to explain and to govern.	Accuracy matters most.	RMSE, calibration	

For a number, error size is the score. RMSE punishes big misses; MAE treats them evenly.

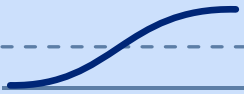
Predicting yes or no

Classification, where the answer is whether something will happen.

EXAMPLES

- Admission
- Churn
- Fraud
- Readmission

Logistic regression is a GLM fitted to a yes or no outcome, the same family as the models on the previous slide.

Logistic regression  Predicts a probability. STRENGTH Transparent. WATCH OUT Limited flexibility.	Decision tree  Creates decision rules. STRENGTH Easy to explain. WATCH OUT High variance.	Random forest  Many trees voting together. STRENGTH Stable, strong performance. WATCH OUT Less interpretable.	Gradient boosting  Sequential tree improvement. STRENGTH Often the highest performance. WATCH OUT Hardest to explain.
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HOW ANY OF THEM IS EVALUATED

AUC for ranking, precision and recall for the trade-off between false alarms and misses, calibration for whether the probability can be trusted.

Pick the measure from the cost of the mistake, not from the model.

Finding groups rather than predictions

Unsupervised methods, used when there is no outcome to predict.

WHEN THIS APPLIES

No target outcome exists.

Nothing is being predicted. The goal is to find natural groupings that already sit in the data.

WHERE WE USE IT

Member segmentation

Provider segmentation

COMMON METHODS

K-means

Choose how many groups you want. Each member is assigned to the nearest group center.

Hierarchical clustering

Builds a tree of nested groups, so you can decide how fine the split should be afterwards.

There is no accuracy score here. Groups are judged by whether they are stable and whether they are useful.

Clustering describes who is in the population. It does not tell you what happens next.

How we open the black box

Three interpretation methods, each answering a different question.

Variable importance THE QUESTION IT ANSWERS <i>Which variables matter most?</i> Ranks the drivers across the whole model. A global view. GLOBAL VIEW OF THE MODEL	Partial dependence THE QUESTION IT ANSWERS <i>How does risk change as a variable changes?</i> Shows the shape of one variable's effect, holding the rest steady. ONE INDIVIDUAL PREDICTION	SHAP values THE QUESTION IT ANSWERS <i>Why did this individual prediction occur?</i> Attributes one person's score to the factors that moved it. A local view.
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SHAP is widely used in practice but sits outside the certificate curriculum; variable importance and partial dependence are the methods covered there.

Explainability is not one thing. Ask which question you need answered before choosing the method.

Three ways to combine models

Ensembles differ in what they are trying to fix.

Bagging

WHAT IT FIXES

Reduce variance

Train many models on different samples and average them. Steadier than any single model.



EXAMPLE

Random forest

Boosting

WHAT IT FIXES

Correct prior mistakes

Each model focuses on what the last one got wrong. Powerful, and easier to overfit.



EXAMPLE

XGBoost, GBM

Stacking

WHAT IT FIXES

Combine model outputs

A final model learns how to weigh several different models' predictions.



EXAMPLE

Blended submissions

WHEN TO REACH FOR ONE

When maximum predictive performance is required, and you have the governance capacity to explain and monitor what you built.

Ensembles buy performance. They charge you in explanation.

Appendix C — Common Ways These Projects Fail

Twelve field-tested pitfalls, mapped to the six leadership questions.

Twelve ways predictive analytics projects fail

Every one maps back to one of our six leadership questions. Ask the question, avoid the pitfall.

The pitfall	Which leadership question catches it
1. Begin without the end in mind — no specific goal	Q1 — What decision changes?
2. Define the project on a foundation the data can't support	Q3 — Do we trust the data and know the business?
3. Wait for “perfect” data before starting	Q3 — Do we trust the data and know the business?
4. Skip data QA — garbage in, garbage out	Q3 — Do we trust the data and know the business?
5. Use data from the future — leakage	Q3 — Do we trust the data and know the business?
6. Rush the process; underestimate data prep (60–80% of effort)	Q3 — Do we trust the data and know the business?
7. Start too big — a first project meant to “revolutionize”	Q1 — What decision changes?
8. Ignore subject-matter experts when building the model	Q3 — Do we trust the data and know the business?
9. Assume data owners will cooperate — ignore the politics	Q6 — Who owns monitoring and change?
10. Build it but never deploy it	Q5 — How will the prediction be used?
11. If results look obvious, you didn't need the model	Q2 — Better than what baseline?
12. Don't define the model's job within business constraints	Q2 — Better than what baseline?

Adapted from Robert L. Mitchell, “[12 predictive analytics screw-ups](#),” Computerworld (July 24, 2013).

A close-up photograph of a business meeting. In the foreground, a person's hands are visible; one hand holds a black pen over a document, while the other rests on the paper. The document contains various business charts, including pie charts and bar graphs. In the background, other participants are partially visible, their hands resting on the table. The entire image has a soft, slightly blurred quality, and a semi-transparent dark overlay covers the bottom half. Centered over this overlay is the text "Thank you!" in a clean, white, sans-serif font.

Thank you!

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